

Iris Dataset Exploratory Data Analysis (EDA)

Objective

Analyze the Iris dataset to understand species distribution, feature relationships, and key statistical patterns.

Dataset

- 150 observations
- 5 columns
- 3 species:
 - Setosa
 - Versicolor
 - Virginica

Key Findings

- No missing values were found.
- 3 duplicate records were removed.
- Petal length and petal width have a very strong positive correlation.
- Setosa has significantly smaller petal measurements than the other species.

Conclusion

The Iris dataset was successfully analyzed using exploratory data analysis techniques. The dataset contained no missing values, and 3 duplicate records were removed. Petal length and petal width showed a strong positive correlation. Setosa flowers had noticeably smaller petal measurements compared to Versicolor and Virginica.

```
In [1]: import pandas as pd
```

```
In [2]: df = pd.read_csv('1) iris.csv')
```

```
In [3]: df.head()
```

```
Out[3]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
In [4]: df.shape
```

```
Out[4]: (150, 5)
```

```
In [5]: df.info()
```

```
<class 'pandas.DataFrame'>  
RangeIndex: 150 entries, 0 to 149  
Data columns (total 5 columns):  
#   Column          Non-Null Count  Dtype  
---  -  
0   sepal_length    150 non-null   float64  
1   sepal_width     150 non-null   float64  
2   petal_length    150 non-null   float64  
3   petal_width     150 non-null   float64  
4   species         150 non-null   str  
dtypes: float64(4), str(1)  
memory usage: 6.0 KB
```

```
In [6]: df.isnull().sum()
```

```
Out[6]: sepal_length    0  
        sepal_width    0  
        petal_length    0  
        petal_width    0  
        species        0  
        dtype: int64
```

```
In [7]: df.duplicated().sum()
```

```
Out[7]: np.int64(3)
```

```
In [10]: df= df.drop_duplicates()
```

```
In [11]: df.duplicated().sum()
```

```
Out[11]: np.int64(0)
```

```
In [12]: df.shape
```

```
Out[12]: (147, 5)
```

```
In [13]: df.describe()
```

```
Out[13]:
```

	sepal_length	sepal_width	petal_length	petal_width
count	147.000000	147.000000	147.000000	147.000000
mean	5.856463	3.055782	3.780272	1.208844
std	0.829100	0.437009	1.759111	0.757874
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.400000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

```
In [15]: df['species'].value_counts()
```

```
Out[15]: species
versicolor    50
virginica      49
setosa         48
Name: count, dtype: int64
```

```
In [20]: %pip install --upgrade pip
```

Requirement already satisfied: pip in C:\Users\OMEN LAPTOP 15\AppData\Local\Programs\Python\Python313\Lib\site-packages (26.1.2)
Note: you may need to restart the kernel to use updated packages.

```
In [21]: pip --version
```

pip 26.1.2 from C:\Users\OMEN LAPTOP 15\AppData\Local\Programs\Python\Python313\Lib\site-packages\pip (python 3.13)
Note: you may need to restart the kernel to use updated packages.

```
In [22]: import sys
print(sys.version)
```

3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)]

```
In [23]: df.corr(numeric_only=True)
```

```
Out[23]:
```

	sepal_length	sepal_width	petal_length	petal_width
sepal_length	1.000000	-0.109321	0.871305	0.817058
sepal_width	-0.109321	1.000000	-0.421057	-0.356376
petal_length	0.871305	-0.421057	1.000000	0.961883
petal_width	0.817058	-0.356376	0.961883	1.000000

```
In [24]: df['species'].unique()
```

```
Out[24]: <StringArray>
['setosa', 'versicolor', 'virginica']
Length: 3, dtype: str
```

```
In [25]: df.groupby('species').mean(numeric_only=True)
```

```
Out[25]:
```

	sepal_length	sepal_width	petal_length	petal_width
species				
setosa	5.010417	3.431250	1.462500	0.250000
versicolor	5.936000	2.770000	4.260000	1.326000
virginica	6.604082	2.979592	5.561224	2.028571

```
In [31]: df.groupby('species').agg({
    'sepal_length': ['min', 'max', 'mean'],
    'petal_length': ['min', 'max', 'mean']
})
```

```
Out[31]:
```

	sepal_length			petal_length		
	min	max	mean	min	max	mean
species						
setosa	4.3	5.8	5.010417	1.0	1.9	1.462500
versicolor	4.9	7.0	5.936000	3.0	5.1	4.260000
virginica	4.9	7.9	6.604082	4.5	6.9	5.561224

```
In [34]: df.sort_values('petal_length', ascending=False).head()
```

Out[34]:

	sepal_length	sepal_width	petal_length	petal_width	species
118	7.7	2.6	6.9	2.3	virginica
117	7.7	3.8	6.7	2.2	virginica
122	7.7	2.8	6.7	2.0	virginica
105	7.6	3.0	6.6	2.1	virginica
131	7.9	3.8	6.4	2.0	virginica

In [36]: `df.sort_values('petal_length').head()`

Out[36]:

	sepal_length	sepal_width	petal_length	petal_width	species
22	4.6	3.6	1.0	0.2	setosa
13	4.3	3.0	1.1	0.1	setosa
14	5.8	4.0	1.2	0.2	setosa
35	5.0	3.2	1.2	0.2	setosa
16	5.4	3.9	1.3	0.4	setosa

In [37]: `%pip install matplotlib`

```

Collecting matplotlib
  Downloading matplotlib-3.10.9-cp313-cp313-win_amd64.whl.metadata (52 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
  Downloading contourpy-1.3.3-cp313-cp313-win_amd64.whl.metadata (5.5 kB)
Collecting cyclers>=0.10 (from matplotlib)
  Downloading cyclers-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib)
  Downloading fonttools-4.63.0-cp313-cp313-win_amd64.whl.metadata (121 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Downloading kiwisolver-1.5.0-cp313-cp313-win_amd64.whl.metadata (5.2 kB)
Requirement already satisfied: numpy>=1.23 in C:\Users\OMEN LAPTOP 15\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib) (2.4.6)
Requirement already satisfied: packaging>=20.0 in C:\Users\OMEN LAPTOP 15\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib) (26.2)
Collecting pillow>=8 (from matplotlib)
  Downloading pillow-12.2.0-cp313-cp313-win_amd64.whl.metadata (9.0 kB)
Collecting pyparsing>=3 (from matplotlib)
  Downloading pyparsing-3.3.2-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\OMEN LAPTOP 15\AppData\Local\Programs\Python\Python313\Lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in C:\Users\OMEN LAPTOP 15\AppData\Local\Programs\Python\Python313\Lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Downloading matplotlib-3.10.9-cp313-cp313-win_amd64.whl (8.2 MB)
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```

```
Downloading contourpy-1.3.3-cp313-cp313-win_amd64.whl (226 kB)
Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.63.0-cp313-cp313-win_amd64.whl (2.3 MB)
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Downloading kiwisolver-1.5.0-cp313-cp313-win_amd64.whl (73 kB)
Downloading pillow-12.2.0-cp313-cp313-win_amd64.whl (7.1 MB)
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----- 2.6/7.1 MB 2.8 MB/s eta 0:00:02
----- 3.1/7.1 MB 2.8 MB/s eta 0:00:02
----- 3.9/7.1 MB 2.8 MB/s eta 0:00:02
----- 4.5/7.1 MB 2.8 MB/s eta 0:00:01
----- 5.0/7.1 MB 2.8 MB/s eta 0:00:01
----- 5.5/7.1 MB 2.8 MB/s eta 0:00:01
----- 6.0/7.1 MB 2.8 MB/s eta 0:00:01
----- 6.6/7.1 MB 2.8 MB/s eta 0:00:01
----- 7.1/7.1 MB 2.8 MB/s 0:00:02
Downloading pyparsing-3.3.2-py3-none-any.whl (122 kB)
Installing collected packages: pyparsing, pillow, kiwisolver, fonttools, cycler, contourpy, matplotlib

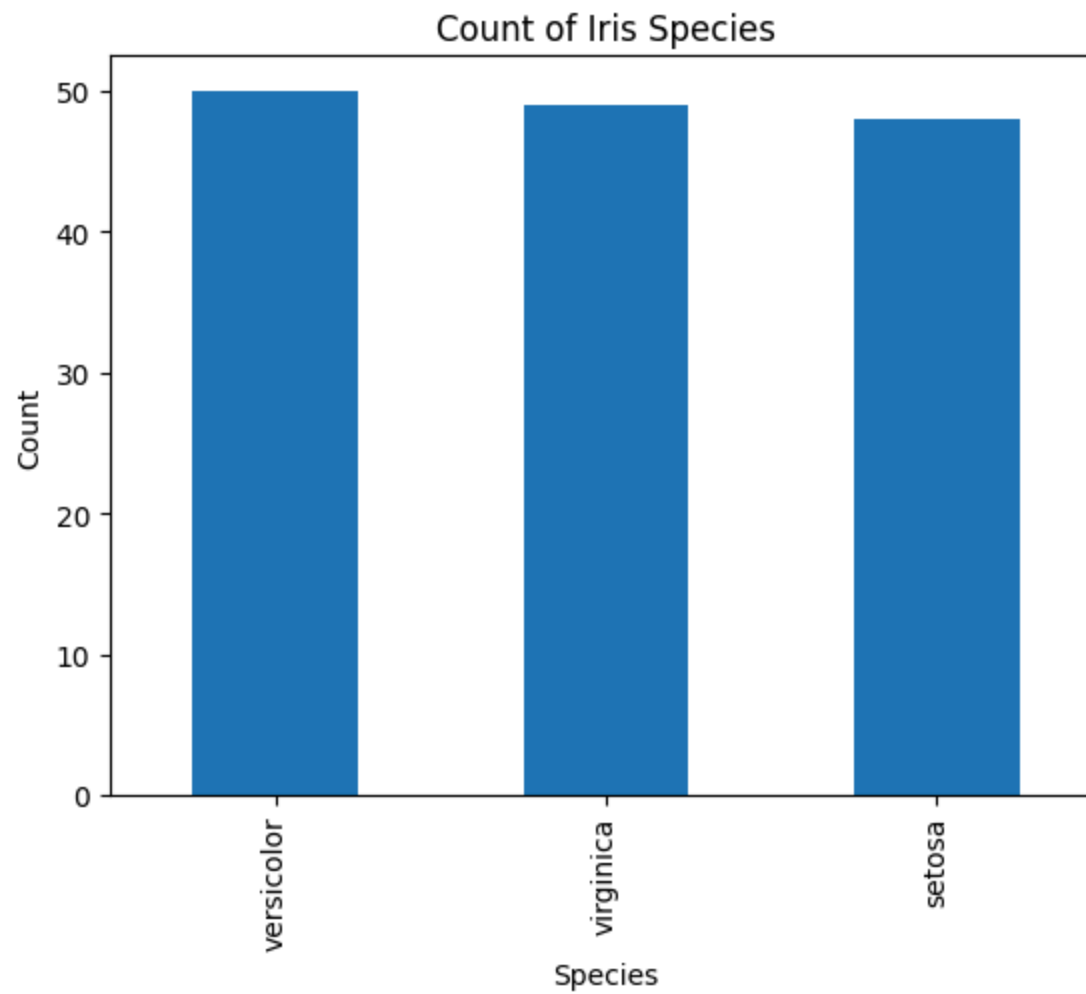
----- 1/7 [pillow]
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```


Successfully installed contourpy-1.3.3 cycycler-0.12.1 fonttools-4.63.0 kiwisolver-1.5.0 matplotlib-3.10.9 pillow-12.2.0 pyparsing-3.3.2

Note: you may need to restart the kernel to use updated packages.

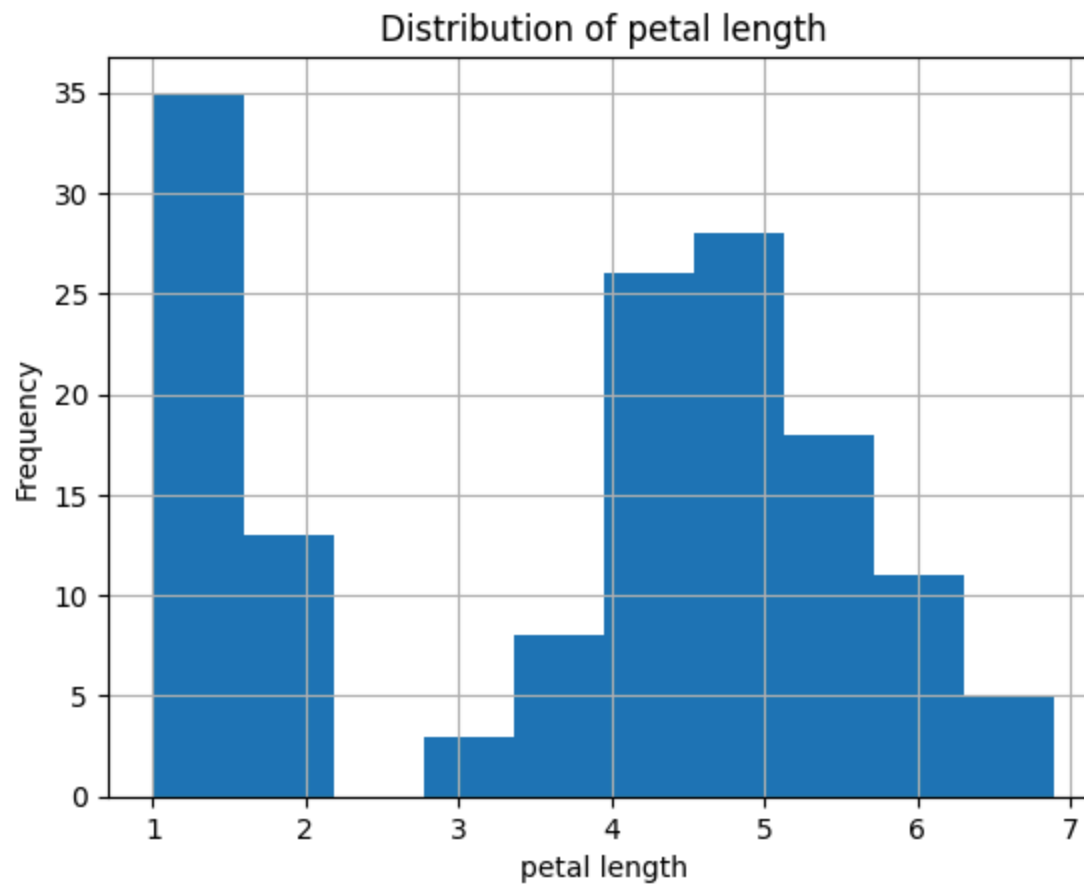
Matplotlib is working!

9/13



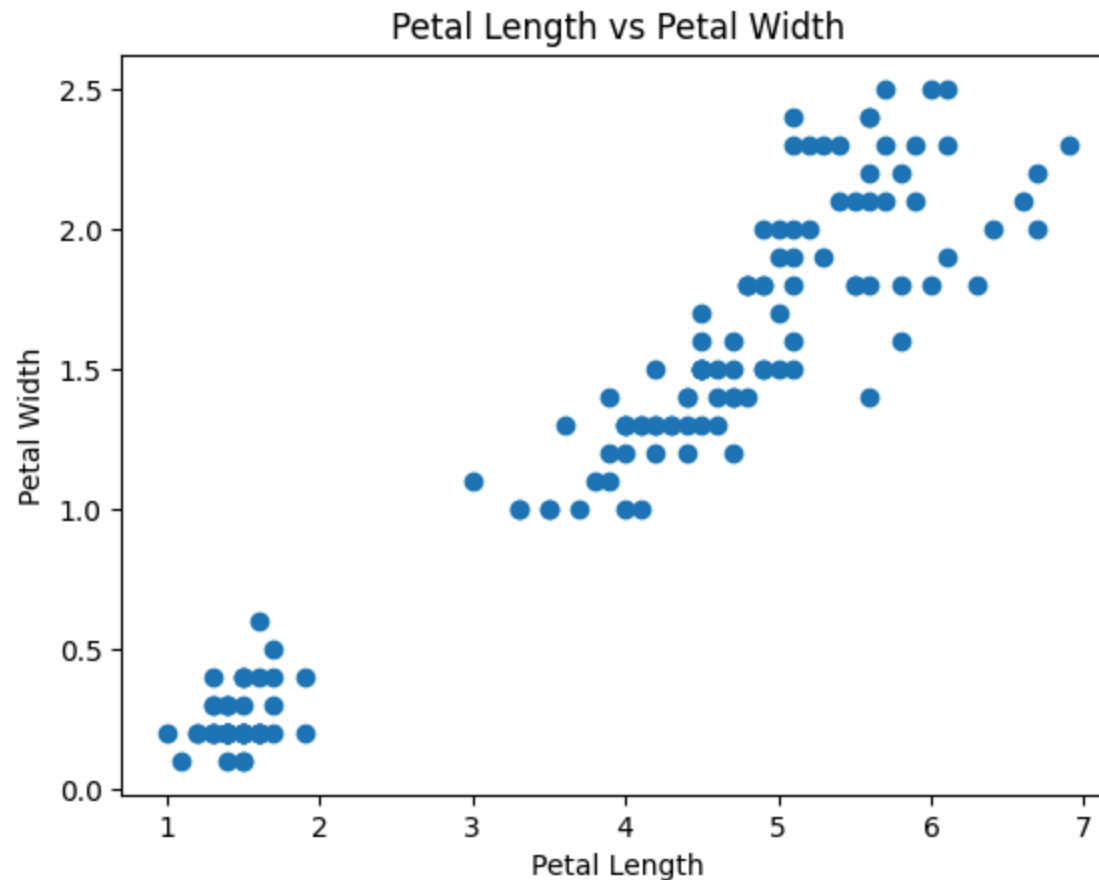
```
In [44]: df['petal_length'].hist(bins=10)

plt.title('Distribution of petal length')
plt.xlabel('petal length')
plt.ylabel('Frequency')
plt.show()
```



```
In [45]: plt.scatter(df['petal_length'], df['petal_width'])

plt.title('Petal Length vs Petal Width')
plt.xlabel('Petal Length')
plt.ylabel('Petal Width')
plt.show()
```



In []: IRIS DATASET EDA REPORT

Objective:

To explore and analyze the Iris dataset using Python and Pandas.

Data Cleaning:

- Dataset initially contained 150 rows and 5 columns.
- No missing values were found.
- Three duplicate records were identified and removed.
- Final dataset contains 147 rows and 5 columns.

Key Findings:

1. The dataset contains three flower species:
 - Setosa

- Versicolor
- Virginica

2. Petal length and petal width have a very strong positive correlation (0.96).
3. Virginica flowers have the highest average petal length and petal width.
4. Setosa flowers have the smallest petal measurements.
5. Histogram analysis shows petal lengths are grouped into distinct ranges, indicating clear differences among species.
6. Scatter plot analysis confirms that larger petal lengths are associated with larger petal widths.

Conclusion:

The Iris dataset is clean, well-structured, and suitable for exploratory analysis and machine learning applications.